

KEM, Russian Federation

WMO: 225220

Lat: 64.950N

Lon: 34.650E

Elev: 26

StdP: 14.68

Time Zone: 3.00 (E03)

Period: 90-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.9	-11.2	-21.3	1.7	-16.5	-15.2	2.4	-10.7	27.2	25.9	24.7	25.1	4.7	290	0.773

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	74.5	60.8	70.2	59.6	66.7	58.4	63.2	70.8	61.1	67.9	59.3	65.3	10.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.1	77.6	66.3	58.0	71.9	64.0	56.4	67.8	62.2	28.6	71.0	27.2	68.1	26.0	65.4	71.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.8	21.0	18.8	-20.9	80.7	8.9	1.4	-27.4	81.7	-32.6	82.5	-37.6	83.3	-44.1	84.3		
(5)			-21.2	67.3	8.9	2.1	-27.6	68.8	-32.8	70.1	-37.7	71.3	-44.2	72.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.7	14.4	15.6	22.7	31.0	39.8	51.8	56.7	55.5	46.7	35.8	24.6	20.7		
	DBStd	17.22	12.86	13.66	9.05	8.59	7.10	7.77	5.69	5.63	6.16	8.01	9.42	9.53		
	HDD50	6146	1104	963	847	570	328	69	8	11	135	440	763	908		
	HDD65	11076	1569	1383	1312	1020	780	405	266	300	551	904	1213	1373		
	CDD50	566	0	0	0	0	13	122	214	181	35	1	0	0		
	CDD65	19	0	0	0	0	0	8	7	4	0	0	0	0		
	CDH74	109	0	0	0	0	1	49	21	37	1	0	0	0		
	CDH80	8	0	0	0	0	0	5	0	3	0	0	0	0		
	WSAvg	9.7	10.7	9.8	10.1	8.9	8.9	9.0	9.0	9.0	9.3	10.6	10.6	10.6		
	PrecAvg	17.1	0.8	0.6	0.6	0.9	1.6	2.2	1.8	2.4	2.1	1.9	1.5	1.0		
(15) Precipitation	PrecMax	27.9	1.7	1.4	1.4	1.9	3.8	4.8	4.3	4.4	4.3	3.3	2.3	2.7		
	PrecMin	10.8	0.2	0.1	0.1	0.0	0.3	0.4	0.1	0.6	0.7	0.2	0.4	0.1		
	PrecStd	3.5	0.4	0.4	0.3	0.5	0.9	1.3	1.1	0.9	1.2	0.8	0.5	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.1	39.6	41.7	56.4	67.8	80.9	77.4	79.0	71.8	54.3	43.1	38.4		
		MCWB	34.9	36.0	36.3	45.9	54.7	62.8	62.8	63.4	60.6	50.6	40.6	35.6		
	2%	DB	35.3	37.1	39.4	49.9	60.9	75.2	72.6	72.9	63.2	51.3	40.3	36.4		
		MCWB	33.3	35.3	35.0	41.9	50.3	60.1	60.6	61.0	55.5	48.3	39.0	34.3		
	5%	DB	33.3	34.4	37.0	45.5	55.3	69.2	68.8	68.4	58.5	49.3	38.5	34.7		
		MCWB	31.6	32.8	33.7	39.7	47.3	59.0	59.5	59.1	54.4	46.8	37.2	33.2		
	10%	DB	31.1	32.3	34.5	42.0	50.8	64.5	65.7	64.7	54.8	46.2	36.4	33.1		
		MCWB	29.6	30.9	32.0	37.8	44.6	57.0	58.7	57.9	51.6	43.9	35.2	31.8		
	0.4%	WB	35.4	36.9	37.8	46.7	56.6	66.4	64.7	64.3	60.3	52.0	41.3	35.9		
		MCDB	37.5	38.2	40.2	55.3	63.9	74.9	72.3	74.7	68.7	53.9	42.3	37.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	33.7	35.1	35.7	42.6	51.9	63.7	62.6	62.4	57.2	49.2	39.2	34.6		
		MCDB	35.4	36.8	38.0	48.2	60.2	70.5	69.3	70.6	61.8	50.7	40.0	35.9		
	5%	WB	32.0	33.1	33.8	40.2	48.0	59.9	60.8	60.3	54.5	46.7	37.4	33.5		
		MCDB	33.3	34.2	36.3	44.7	54.6	68.5	66.5	66.3	57.7	48.8	38.6	34.6		
	10%	WB	29.8	31.0	32.1	37.9	45.2	56.4	59.2	58.6	52.2	44.2	35.4	31.8		
		MCDB	31.0	32.4	34.6	41.7	50.3	64.7	64.7	63.8	54.5	46.2	36.2	32.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	11.2	11.9	11.6	10.7	11.5	10.3	11.0	8.7	7.5	8.6	9.9		
		MCDBR	11.2	8.8	11.9	14.9	20.7	19.8	16.3	18.8	14.8	9.8	7.4	7.9		
		MCWBR	10.3	7.3	9.0	9.2	11.8	10.9	7.9	9.3	8.8	8.2	6.4	7.3		
	5% WB	MCDBR	11.1	8.4	10.8	13.8	19.2	18.0	14.0	15.8	12.8	9.4	7.4	8.0		
		MCWBR	10.3	7.3	9.0	9.3	11.8	11.2	8.0	8.7	8.2	8.5	6.4	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.218	0.280	0.309	0.342	0.354	0.350	0.378	0.355	0.331	0.303	0.346	0.358		
	taud		1.953	2.093	2.152	2.273	2.425	2.483	2.432	2.472	2.518	2.364	2.563	2.613		
	Ebn at Noon		90	187	239	261	271	275	261	257	240	187	87	37		
	Edh at Noon		8	19	29	32	31	30	31	27	21	15	7	4		
(44) All-Sky Solar Radiation	RadAvg		39	221	712	1187	1479	1733	1589	1133	641	233	50	11		
	RadStd		4	36	73	114	129	154	174	116	67	30	8	1		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (8 neighbors)	+1.15	N/A	N/A	N/A	N/A	N/A	N/A	-409	N/A		

Nomenclature: See separate page